

IND 598 - Advanced UX/UI Design- UX for Emerging Technologies

Mercure - Healthcare Management Application

Final Report

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1. Introduction

Managing medical information is something almost everyone deals with, but it is rarely easy. Parents juggling vaccines, prescriptions, and school health forms. Caregivers who are trying to track a dementia patient's daily medications. Young adults who can never find that old lab report at the doctor's office. The problem is not that people do not care about their health. The tools available to manage it are scattered, confusing, and stressful to use.

Our project, Mercure, was designed to fix exactly that. We built a mobile healthcare management application that acts as a single, secure home for all your medical information, including documents, medicines, appointments, and even emergency contacts. The goal was simple: reduce the mental load of managing health information so users can focus on what actually matters, which is staying healthy.

The Problem We Are Solving

Right now, people store medical information across paper files, WhatsApp messages, phone gallery photos, sticky notes, and calendar reminders. When they need a document quickly at a clinic, during an emergency, or at a school nurse's office, they often cannot find it. This causes real harm: missed doses, repeated tests, delayed care, and a lot of unnecessary anxiety. The fragmentation is especially bad for primary caregivers who manage healthcare for children or elderly relatives because they are tracking the information of multiple people at once.

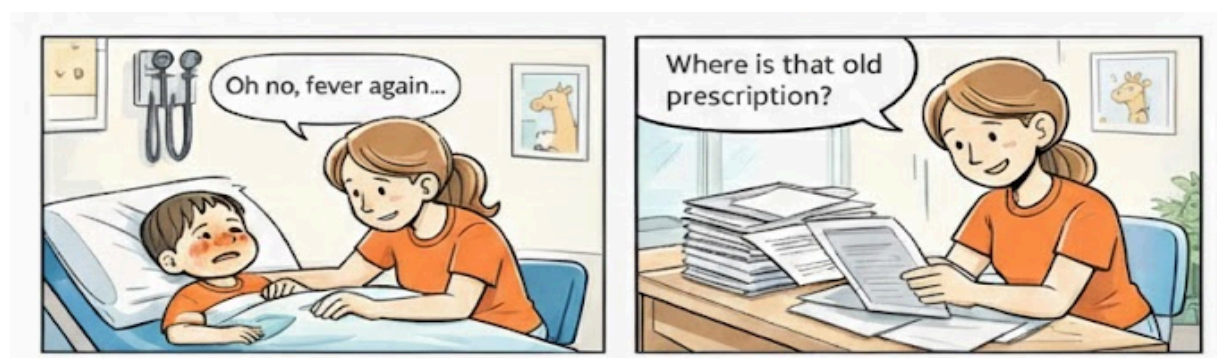




Figure 01: A parent struggling to find a prescription during a doctor visit.

Our Solution

Mercure brings all of this into one place. The app lets users upload and organize medical documents by category (lab reports, prescriptions, X-rays), set medication reminders with dosage details and visual medicine images, track appointments on a timeline, and access an emergency SOS feature that connects to saved contacts and allows location sharing. The design is built around simplicity and privacy. Users should feel calm when they open the app, not overwhelmed.

Key Findings

Our most striking finding from prototype evaluation was that the appointment timeline and medicine card views were immediately understood by users, with one participant describing the concept as very useful and well-structured. However, we also discovered that navigation between sections like Profile and Documents caused significant confusion, and that privacy on the home screen, like allergies being visible when others are nearby, was a concern participants raised unprompted. These insights drove major changes from our low-fidelity to high-fidelity prototype.

2. Background

Related Literature

Research in health informatics has long highlighted the problem of fragmented patient records. Studies on patient-centered care emphasize that when individuals have easy access to their own health information, outcomes improve. They are more likely to adhere to medication schedules, show up for follow-up appointments, and communicate clearly with care providers. The concept of a Personal Health Record (PHR) has been studied extensively, with findings showing that users want these systems to be simple, trustworthy, and private.

Work on cognitive load theory is also relevant here. When caregivers have to mentally track multiple medications, appointments, and documents across different systems, their cognitive load increases significantly, and so do errors. Design principles like chunking information, using visual recognition over recall, and providing clear feedback all come from this body of research, and we applied them throughout our design process. Norman's principles of affordances, mappings, and feedback guided how we annotated our wireframes and structured our screens.

Existing Solutions

There are a few apps that touch on this space. Apple Health aggregates some health data from devices, but it does not support document uploads or caregiver-specific workflows. One of our test participants actually referenced it as a benchmark, which tells us users are aware of it but still feel like something is missing. MyChart, offered by many hospital systems, is doctor-facing rather than patient-driven and is tied to a specific health system, so it does not work across providers. Apps like Medisafe focus narrowly on medication reminders but do not handle documents, appointments, or emergency features.

Mercure's value comes from combining all of these needs into one experience, including documents, medications, appointments, and emergencies, and designing it specifically around caregivers and patients rather than providers. That focus on the non-clinical user who is managing information rather than entering it is what makes our approach different.

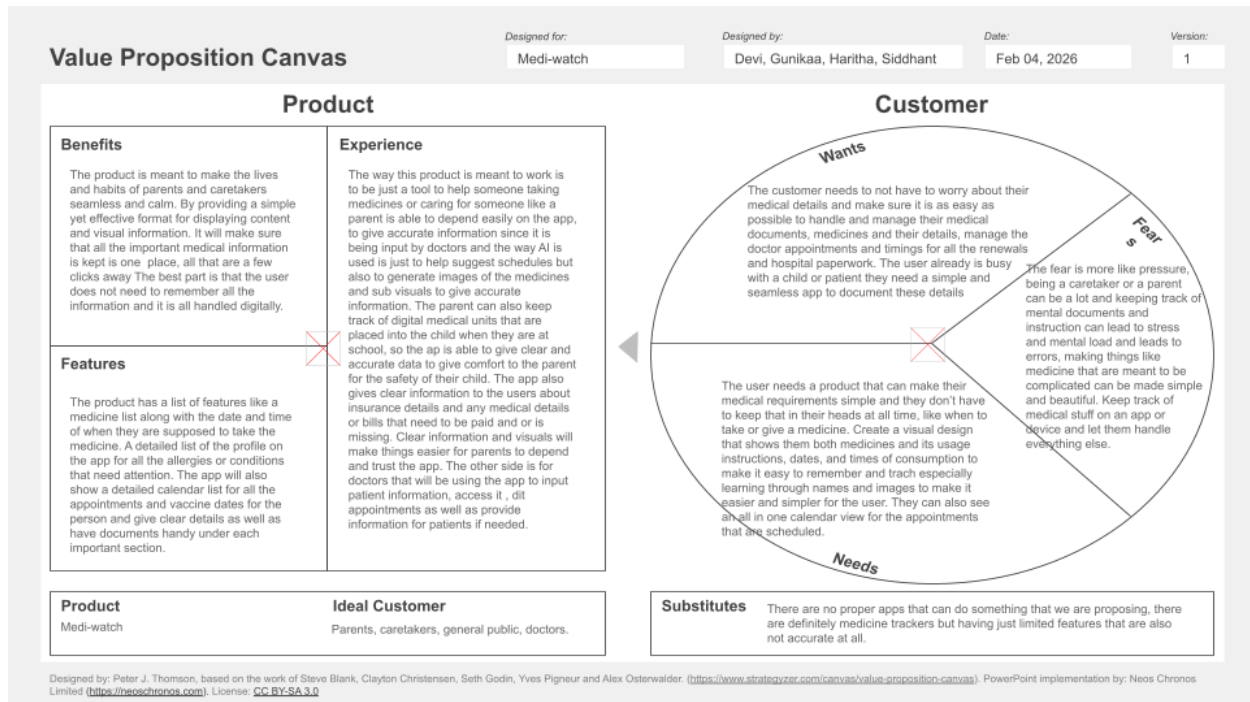


Figure 02: Mapping user needs, pain points, and opportunities to define a focused and meaningful solution.

3. Context Study

Understanding Our Users

We created two personas to represent our main user groups. The first is the Primary Caregiver, someone like a 35-year-old teacher who manages healthcare for her children and acts as the link between the clinic, school nurse, pharmacy, and insurance provider. Her biggest pain points are fragmented information, fear of forgetting something critical, and the communication overhead of having to explain the same medical history repeatedly. She wants everything in one place and needs it to be shareable quickly. The second persona is the Medical Assistant, a 30-year-old nurse practitioner who works in a pediatric clinic under time pressure. She needs accurate, structured information from parents before she enters the exam room, and gets frustrated when records are incomplete, inconsistent, or hard to verify.

Our empathy map for the primary caregiver was organized around four quadrants. In "Says," users expressed things like "I keep forgetting which medicine to give at what time" and "I need

all my kids' medical information in one place." In "Thinks," they worried about what would happen if their phone died and they lost access to medical details. In "Does," they create child profiles, upload documents, and share app access with spouses. In "Feels," the map captures anxiety about fragmented information but also relief and calm when things are centralized. This emotional dimension shaped our design goal of not just organizing information but making users feel in control.

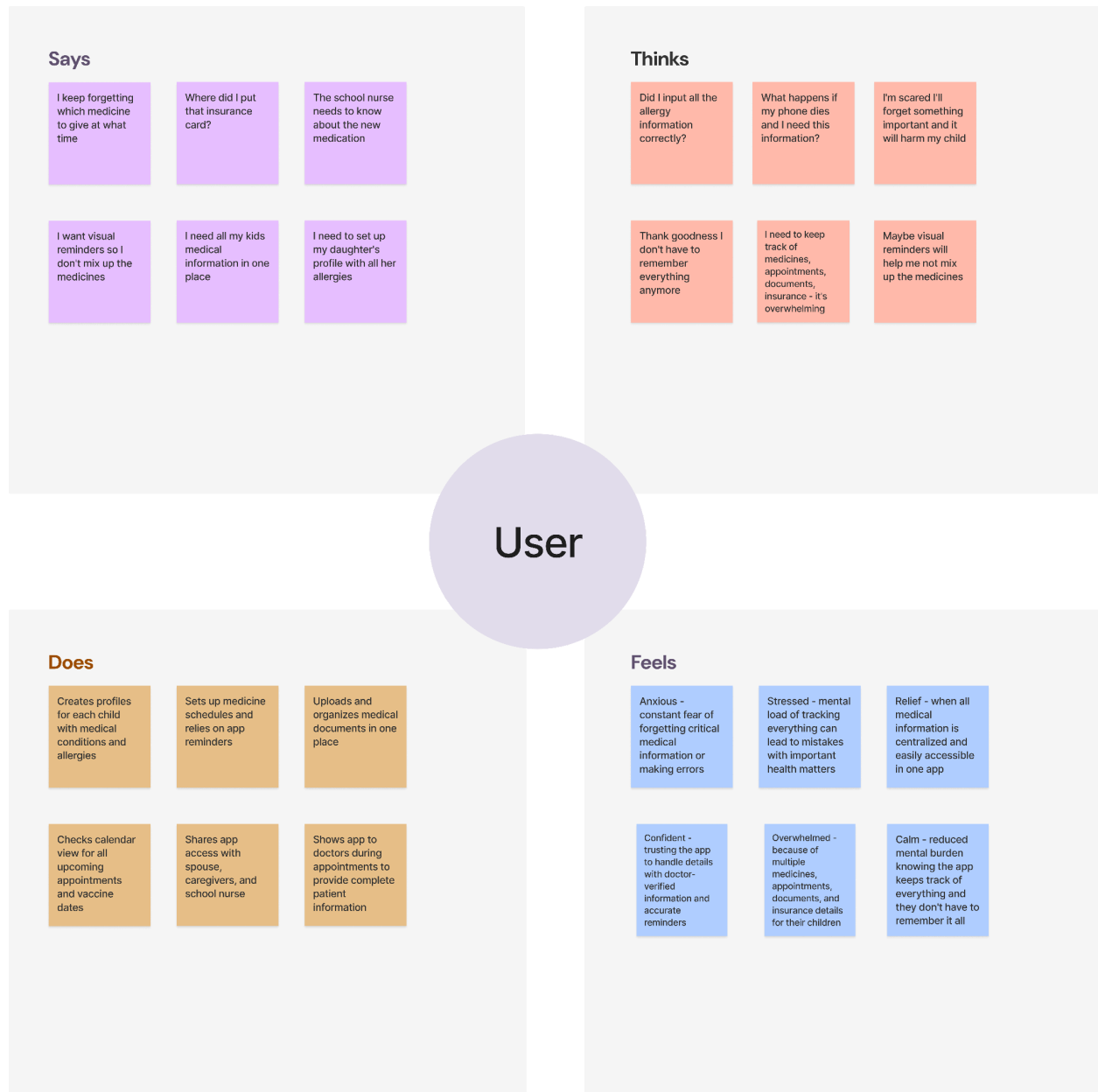


Figure 03: Understanding user thoughts, emotions, and behaviors to design for both functional and emotional needs.

Exploratory Study - Interviews

We ran semi-structured interviews with four people, one mock interview with a classmate, and three external interviews with people aged 22 to 26. We chose open-ended questions to understand not just what people do, but how they feel about it. Some key questions included: How do you track medical information? Does it feel stressful? What is the most frustrating part?

Even though our external participants were young adults managing their own health rather than dependents, the same patterns came up: scattered records, stress about medical terminology, difficulty finding old reports, and a strong interest in a digital solution as long as it was private and secure. Our mock interview participant specifically mentioned wanting precise alarm-based reminders and highlighted the importance of privacy between patient and doctor. That finding directly influenced our decision to include privacy-safe notifications in the final design, which are reminders that do not show sensitive details on the lock screen.

Affinity Diagrams and Key Insights

We built two affinity diagrams, one from video observations and one from our healthcare interviews. For the interview-based diagram, we organized insights into Activities, Environments, Interactions, Objects, and Users. The most important themes that emerged were: reliance on fragmented systems (paper files, WhatsApp, gallery photos), repetition of medical history at every visit as a major frustration, trust and privacy as non-negotiable requirements, and a strong desire for a simple and minimal interface rather than a feature-heavy one. These themes became the foundation of our POV statements and our "How Might We" questions, for example, "How might we centralize medical information so caregivers can access it quickly during appointments and emergencies?" These HMW questions guided every design decision that followed.

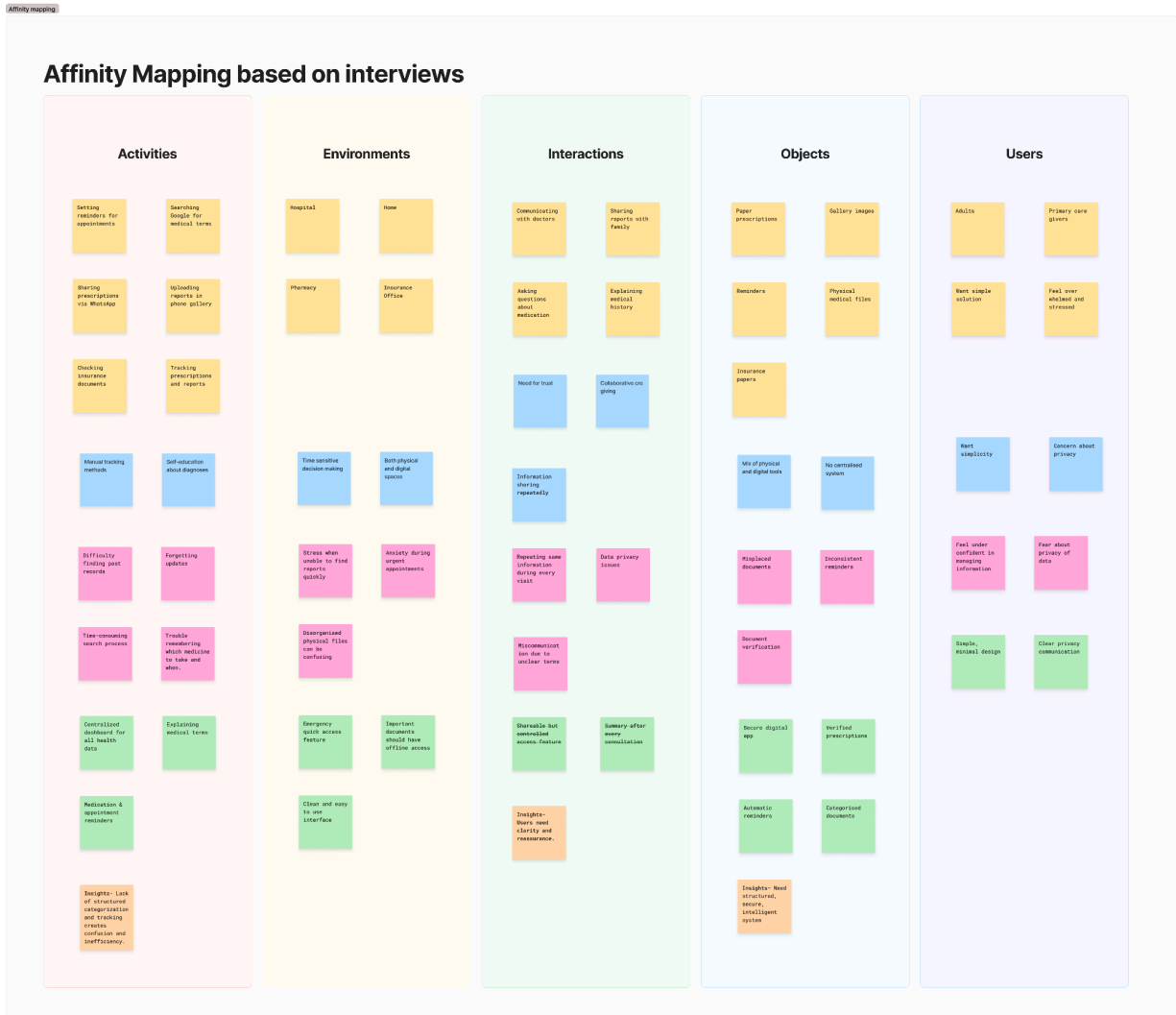


Figure 04: Organizing research data into patterns to uncover key themes like fragmentation, trust, and usability.

4. Design

Design Alternatives - Storyboards and Crazy 8

Each team member wrote an epic and created a storyboard for their assigned user group. Storyboard 1 follows a parent whose child gets a fever and who, thanks to the app, can pull up the prescription instantly at the doctor's office with no paper-digging required. Storyboard 2 showed a caregiver managing a dementia patient's complex medication and diet schedule.

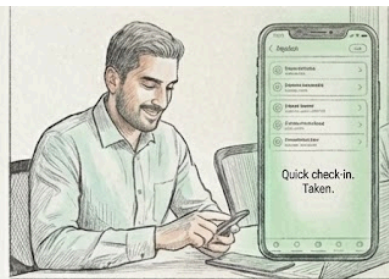
Storyboard 3 was about a working professional who wanted to manage his blood pressure medication discreetly in a shared office, without his condition appearing on his lock screen. Storyboard 4 showed a doctor using the app to keep patient records and appointments organized.



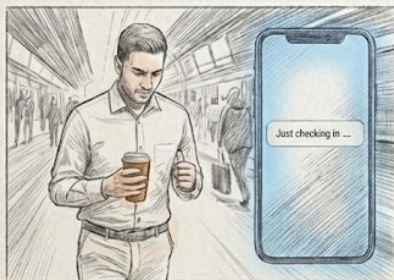
The Public Interruption



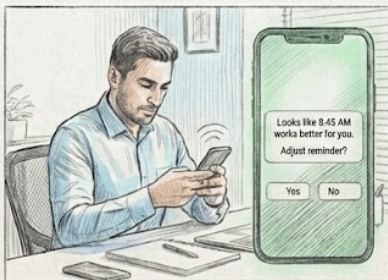
The Uncertainty Trap



The Discreet Confirmation



The Gentle Nudge



The Adaptive Shift



The Reward of Consistency

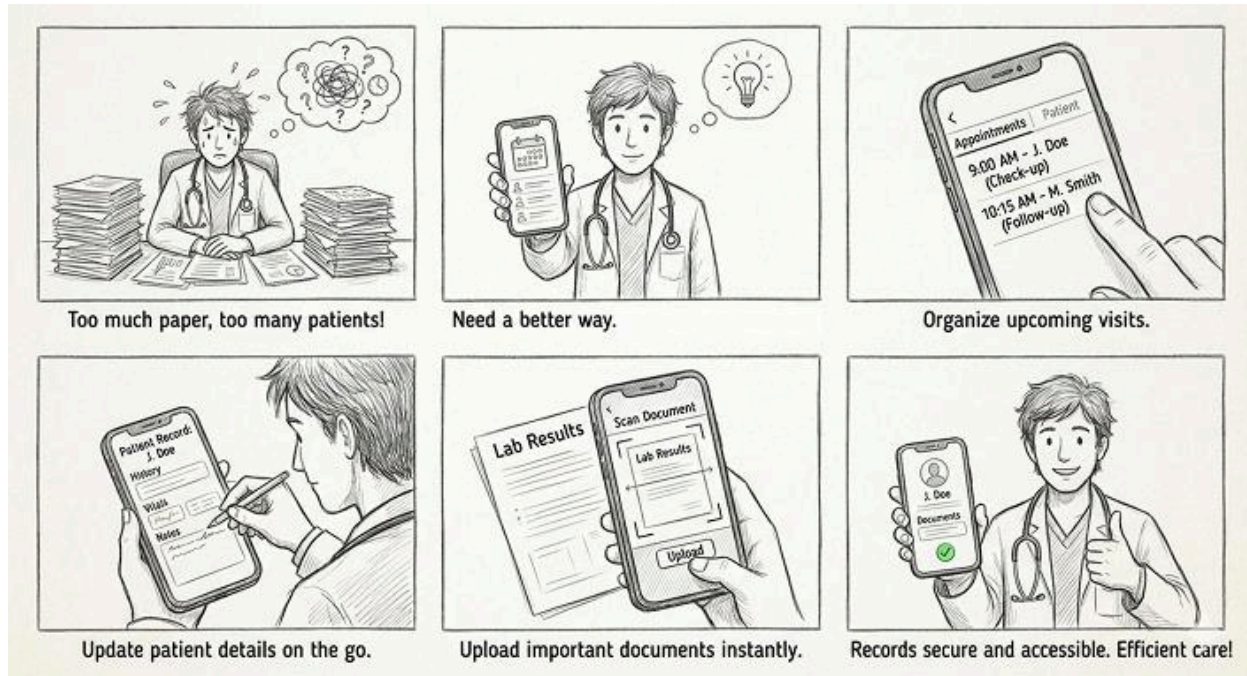


Figure 05: Exploring different real-life scenarios to guide feature design and user-centered decision making.

For our Crazy 8 ideation session, each member sketched eight layout variations for their feature in eight minutes. After reviewing the sketches together, we voted on the strongest ideas. For Feature 1 (Centralized Documentation), we chose the dashboard-style layout with large category icons, including Reports, Medicines, Appointments, Allergies, and History, because it matched users' mental models of organizing by category rather than file type, reduced cognitive load, and worked well for fast access during a doctor visit. For Feature 2 (Reminders), the chosen sketch combined a checklist view organized by morning, afternoon, and night with checkboxes and dosage details. For Feature 3 (Timeline), we merged several sketches into a combined design. For Feature 4 (Emergency SOS), we explored concepts that prioritized the 911 button as the most visually dominant element with the shortest interaction path.

Design Iteration

After sketching, we built detailed wireframes with content and control annotations. We documented exactly what each element does and why it is there. For example, the Upload button uses a plus icon as a strong signifier placed in the thumb-friendly area of the screen, and the category tabs are always visible to improve discoverability. We also built user flow diagrams for each feature to make sure the interaction sequences were logical and complete. Feedback from

classmates during a peer review session pushed us to improve navigation clarity and reconsider how appointment notes were accessed. These issues reappeared in later evaluation rounds and confirmed that the peer feedback was accurate.

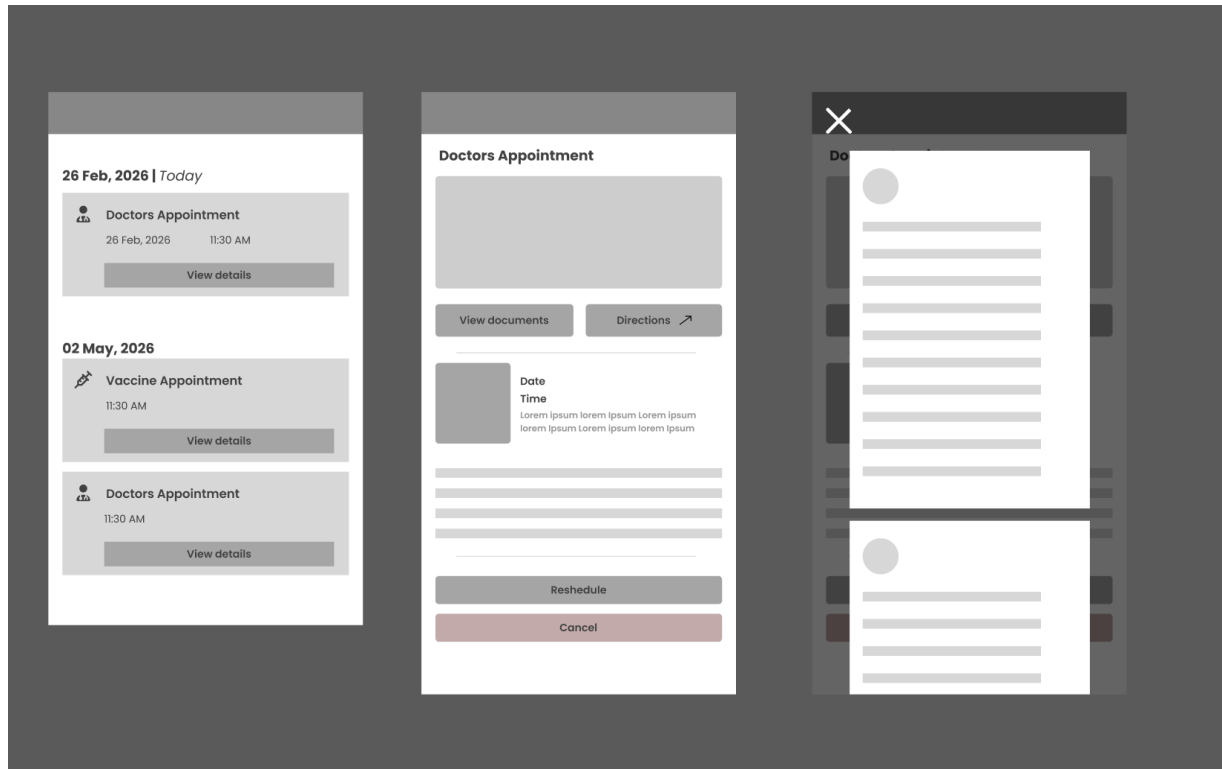


Figure 06: Early layout explorations focused on structure, navigation, and usability before visual refinement.

Final Design

The final high-fidelity prototype was built in Figma with a full design system. The color palette is centered on blue, chosen for trust and professionalism, with green used for confirmed or verified states, and red reserved for warnings only. Typography uses DM Sans and DM Serif Display at sizes ranging from 14 to 24 points, giving a clear hierarchy between headings, subheadings, and body content. Spacing follows an 8-point grid with 16px margins throughout.

The app has four main sections accessible from a bottom navigation bar: Home, Profiles, Timeline, and Medicines. The Profiles screen centralizes each patient's documents, appointments, and medical summary. The Timeline shows appointments in chronological groups with date headers and clear "View Details" buttons. The Medicines screen uses a card-based layout with medicine name, dosage, issue date, and a green "Verified" badge for quick scanning.

The Emergency screen has a visually dominant 911 button at the bottom, emergency contact management, voice note capability, and location sharing, all in the fewest possible taps.

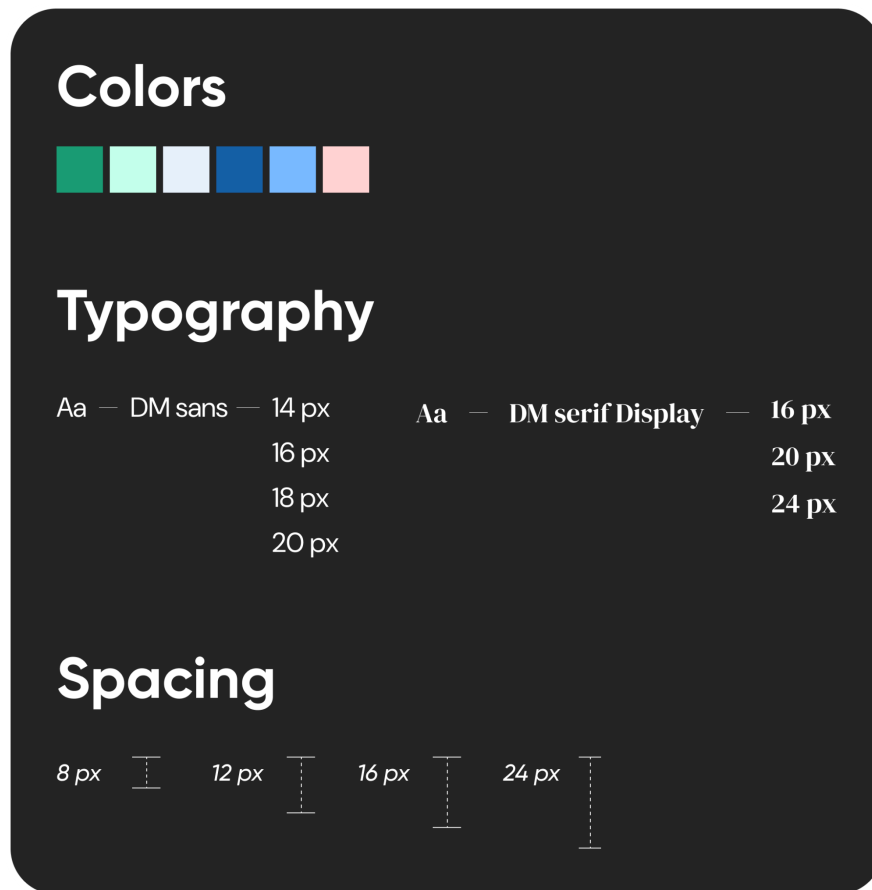


Figure 07: Design System

5. Implementation and Test

Low-Fidelity Prototype and Evaluation

We created two types of low-fidelity prototypes in parallel. One part of the team built a paper prototype using the Wizard of Oz method, where one team member acted as the "computer" and swapped paper screens based on what the participant tapped. The other part built interactive Figma wireframes for the same features. We focused on two core features for this round: Centralized Documentation with Medicine Storage, and Reminders with Dosage Tracking and Notifications.

We tested with one classmate (mock evaluation) and two external participants. The mock evaluation gave us quick, early feedback. The red icon on one screen did not communicate its purpose, the bottom navigation text was too small, and users tried to swipe sections that were not scrollable. The participant also recommended adding appointment distance information and color-coded medication indicators, and she referenced Apple Health as a mental model, confirming we needed to meet familiar design expectations.

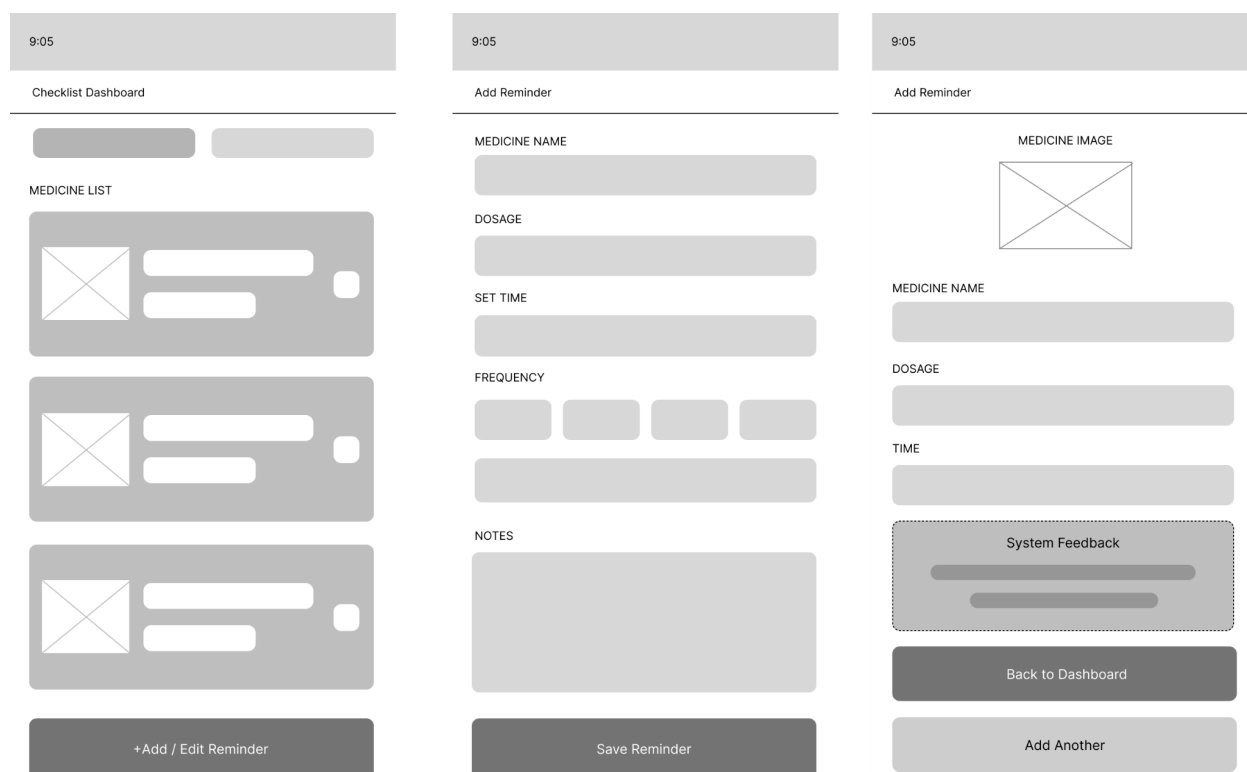


Figure 08: Low-Fidelity wireframes

Participant 1 in the external evaluation found the appointment timeline intuitive and well-structured, but struggled with the reminder entry form when adding multiple dosages, as the input flow was unclear. She also misidentified a section element as a banner rather than a clickable button, and raised a strong privacy concern: she did not want allergies or symptoms visible on the home screen when other people were around. Participant 2 understood the document organization and medicine checklist well, but wanted clearer grouping labels for dosage timing across morning, afternoon, and night. She also suggested that appointment notes should be more directly accessible rather than buried under profile documents.

High-Fidelity Prototype and Evaluation

For the high-fidelity round, we moved to a full-color Figma prototype with a consistent design system. We addressed the low-fidelity feedback by improving navigation labels, increasing font sizes, adding better visual contrast, and clarifying the button hierarchy. We conducted two rounds of evaluation: a heuristic evaluation with two classmates using Nielsen's heuristics, and a cognitive walkthrough with one external participant across three tasks.

The heuristic evaluation identified six usability issues. The most critical were inconsistent screen layout structure between Profile and Appointment pages (making them feel like different apps), a cramped appointment detail page, weak visual hierarchy, and oversized navigation elements. These were all rated major severity because they directly affected readability, navigation efficiency, and accessibility.

The cognitive walkthrough covered three tasks: viewing today's medicines, checking appointment details, and finding appointment notes. The first two tasks went smoothly. The home screen was intuitive, and tapping an appointment card to see details felt natural. The biggest failure was Task 3. The participant went directly to the Timeline section to find appointment notes, which is exactly what you would expect someone to do. But the notes were stored under Profile > Documents instead, which caused considerable confusion. This confirmed a mental model mismatch we had seen hints of in the low-fidelity round. On the positive side, the SOS button was noticed immediately and understood without any explanation, which was a clear design win.

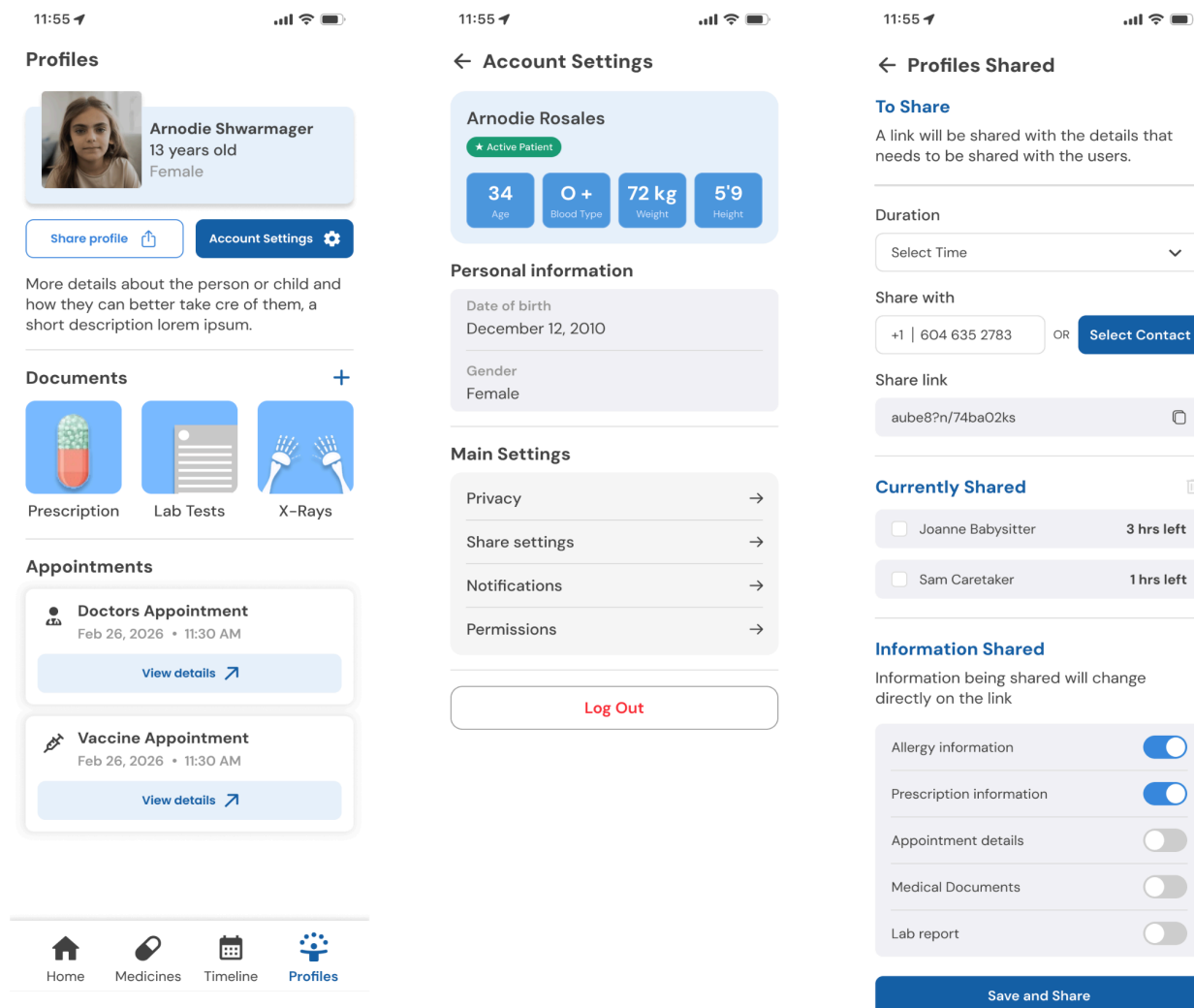


Figure 09: High-Fidelity wireframes of the profile and profile sharing screens

Design Changes From Feedback

Based on all evaluation rounds, we identified the following as the most important improvements to make: adding a direct "View Notes" link on the appointment details screen to avoid the Profile > Documents path; renaming the Documents section to something like "My Records and Notes" so users understand what is inside before tapping; updating the "Timeline" label to better signal both past and upcoming appointments; improving spacing and contrast on the appointment page; and keeping sensitive health details like allergies hidden by default on the home screen to address the privacy concern multiple participants raised independently.

6. Conclusion

Reflections on the Design Process

The biggest lesson from this project was that user testing catches things that no amount of internal review will find. We thought the Profile section was a logical place to store documents because that is where a patient's information lives. But users naturally went to the Timeline or Appointments section instead, because that is where they think about their health events. The mismatch was invisible to us until we watched someone struggle with it in real time. That experience reinforced why testing early and iterating often is so important in UX design.

Another key insight was about the emotional dimension of health apps. Multiple participants mentioned feeling anxious when opening an app that deals with medical information. That is a real phenomenon, and it means every design decision, including color, font size, how you word an error message, and whether sensitive information is visible on the home screen, carries more weight than it would in a productivity app. Designing for emotional reassurance is not just a nice-to-have. It is a core requirement in this space.

We also learned that privacy is a design constraint, not just a legal one. When participants asked unprompted for allergies to be hidden on the home screen, or wanted notifications that did not reveal the medication name on a shared lock screen, they were telling us that the social context of using a health app matters. Future designers working on similar products should treat privacy not as a settings menu afterthought but as a first-class design requirement that shapes the entire information architecture.

Current Limitations and Future Plans

Our current prototype has some clear gaps. The appointment notes path remains unintuitive and needs to be redesigned. The dosage entry form for multiple times-per-day medications is still confusing and needs more user testing to get right. We also did not fully implement the Emergency SOS and Messaging features in the high-fidelity prototype, so those remain at the wireframe stage. The Privacy First smart notification system from our Epic 3, where notifications adapt to the user's environment, is a compelling feature that we ideated on but did not prototype in detail.

If we were to continue developing Mercure, the next steps would be to run another round of usability testing with the revised navigation structure, implement the full emergency flow with

live location sharing, and explore how AI could help simplify medical terminology in uploaded documents, which was one of the most consistent pain points we heard from users across every round of research. We would also want to expand testing to older adults and non-English speakers, since our current sample skewed young and tech-literate, which means we may be missing important accessibility and language needs.